CCS6334
DIGITAL & COMPUTER FORENSICS

Lecture 5:

Digital Forensic Laboratories and Tools

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Learning Objectives

After studying this chapter, you should be able to:

- Explain the functions and importance of digital forensic laboratories.
- Describe the purpose and roles of digital forensic laboratories in investigative processes.
- Identify key laboratory requirements and best practices for setting up and operating a digital forensic lab.
- Discuss standard forensic processes and evaluate various forensic tools used in digital investigations.

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Digital Forensics Laboratories

- **Growing demand**: Rapid expansion of training & education in government and private sectors.
- **Employment**: Many new practitioners join forensic labs conducting digital examinations.
- **Challenges**:
  - Budget constraints → reduced staffing & specialist training.
  - Rising caseloads → risk to quality of work.
  - High costs → equipment, personnel, infrastructure.
- **Solutions**:
  - Knowledge sharing across community → guidelines, automation, faster analysis of large datasets.
- **Business context**:
  - Corporate failures often involve fraud & asset theft.
  - Forensic auditors rely on practitioners to analyze digital evidence.
  - Private labs face heavier caseloads → need more qualified staff.
- **Takeaway**: Digital forensic labs are vital but resource-intensive; collaboration, automation, and skilled staffing are essential to meet rising demands.

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Purpose of Digital Forensics Laboratories

- **Limited availability**: Few organizations have labs; costly equipment & skilled personnel are scarce.
- **Integration**: Often part of security-incident response programs.
- **Established labs**: Found in law enforcement, defense, intelligence, large financial institutions, and global accounting firms (e.g., Deloitte, EY, KPMG, PwC).
- **Smaller organizations**: Rely on private practitioners or accounting firms; high costs limit access for many defendants.
  - Core functions:
  - Provide essential support & coordination for practitioners.
  - Enhance standards to meet court expectations.
  - Improve efficiency in case processing & resource management.
- **Critical need**: Formal, well-designed labs ensure compliance with evidentiary rules; without them, examinations risk being fragmented and ineffective.

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Digital Forensics Laboratories in Malaysia

Government Digital Forensic Labs

- MyDFLAB (Malaysia Government Digital Forensic Lab): Located in Cyberjaya, this lab is part of the Jabatan Digital Negara (National Digital Department) and provides digital forensic services for government entities.
- CyberSecurity Malaysia (CSM): An agency under the Digital Ministry, CSM has a comprehensive digital forensics laboratory with capabilities in various areas, including computer, multimedia, mobile phone, and data recovery.
- Royal Malaysia Police (PDRM): The police force has its own digital forensics laboratory, as seen from the clients listed by CyberSecurity Malaysia.
- Malaysian Anti-Corruption Commission (MACC): The MACC has a digital forensics technology division.

Private Digital Forensic Labs

- LGMS Berhad, Siaga Informatics, OG IT Forensics Services, PwC Malaysia, etc

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Accreditation of Digital Forensic Laboratories

- Accuracy is critical → faulty examinations risk wrongful convictions or unsafe trials.
- Accreditation raises standards → builds court confidence in forensic analysis & evidence handling.
- Current situation:
  - Some jurisdictions require accreditation, most do not.
  - Calls for practitioner certification with proficiency testing for expert testimony.
- Challenges:
  - Global consensus on accreditation unlikely.
  - Practitioners often must prove credentials individually in court.
- US leadership:
  - Daubert Test widely used to validate tools & practitioner competence.
  - Driven by case law and precedent.
- International standards:
  - Expectation of ISO 17025 accreditation or US ASCLD/LAB equivalent.
  - ASCLD/LAB labs meet ~360 ISO standards; fewer applied to digital forensics labs.

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Key Guarantees Required from Forensic Labs:

- Appropriate safe custody of physical exhibits
- Validation of the forensic processes and tools used
- Adherence to forensic best practices
- Forensic computers being in effective working order
- Verifiable calibration of forensic tools

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Best practices for digital forensics laboratories

- **Governance & Case Management**
  - Manuals, regulations, and oversight ensure professionalism and smooth case handling
  - Examinations and reports cross-checked for accuracy, completeness, and anomalies
- **Custody of Evidence**
  - Evidence register tracks all movements and personnel access
  - Proper tagging & cross-referencing of devices/media required
- **Tool & Examination Standards**
  - Proficiency testing of forensic software/tools before use
  - Imaging/copying of digital media must follow custody & reporting protocols
- **Archiving & Disposal**
  - Cataloging and secure storage for appeals, cold cases, and investigations
  - Disposal/destruction governed by legislation after retention period

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Physical Security of DF Laboratories

- **Evidence Protection**
  - Prevents contamination, tampering, and unauthorized access
  - Chain of custody must be strictly maintained
  - Use security-grade lockers, cabinets, or safes with dual locks
- **Workstation & Lab Access**
  - Forensic equipment may run after hours — secure workspaces are essential
  - Implement strict access control, isolation, and alarm systems
  - Supervise all visitors to prevent interference
- **Perimeter & Privacy Safeguards**
  - Control lab entry and segregate internal workstations
  - Shield monitors from external view to protect privacy

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Network & Electronic Requirements of DFL

Electrical Infrastructure

- Approved wiring protects sensitive equipment from power surges, Isolated circuits (≤2 terminals) reduce outages and enhance data security.
- Use surge protectors, UPS, and simplified cable management

Workstation Setup

- Each examiner requires a separate workstation and storage space for exhibits being currently examined
- Network access required, but must be shielded from interception
- Secure storage for active exhibits

Electromagnetic Protection

- Disarm Wi-Fi/Bluetooth devices; remove SIMs; enable flight mode before analysis
- Use Faraday bags or screened rooms to block interference

Environmental Controls

- Air conditioning, antistatic carpeting, and acoustic soundproofing ensure safety, privacy, and equipment longevity

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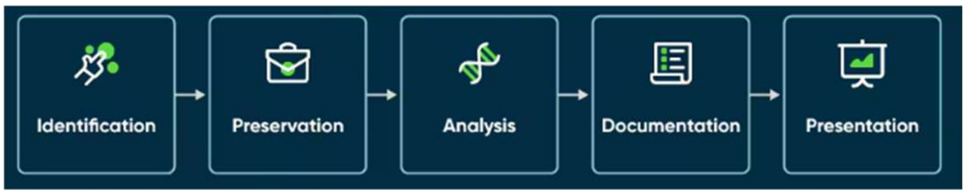
EMERGING PROBLEMS IN DIGITAL FORENSIC



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Data Storage Challenges

Persistent Storage Problem

- Cataloging & archiving evidence is costly and time-consuming
- Legacy tools risk redundancy, licensing issues if vendors cease operations

Large Dataset Complexity

- Processing requires advanced expertise, often misunderstood in legal settings
- Current tools are time-consuming → urgent need for research solutions

Criminal vs Civil Examinations

- Both seek similar evidence, though standards differ
- E-discovery approaches should be adapted for digital forensics

Impact on Legal Hearings

- Storage & analysis challenges can delay or compromise proceedings

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The Myth of Forensic Imaging

- **Court Expectations**: Evidence must be forensically sound: unmodified, uncontaminated, and properly logged. No universal requirement for full-drive imaging
- **Legal Framework**: US Federal Rules of Evidence §901(a): authentication requires proof of recovery process. Over-collection risks privacy breaches and irrelevant data inclusion
- **Selective Recovery vs Full Imaging**: Imaging recovers all data (including deleted/corrupted) → costly, time-consuming, often low evidentiary value. Courts increasingly favor targeted recovery over “bucket” full-drive imaging
- **Use Cases**: Imaging still preferred for criminal/corporate cases involving hidden or deliberately deleted data. Ineffective for cloud-based sources (e.g., email, Dropbox)
- **Challenges**: Large dataset sizes → resource-heavy, slow, expensive. 90% of civil cases: imaging deemed excessive and unjustified

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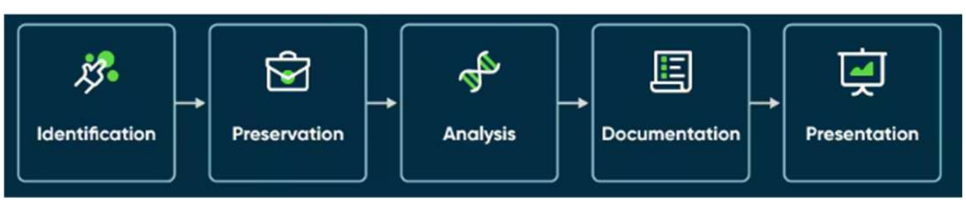
PROCESS AND FORENSICS TOOLS



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E-Discovery: Evidence Recovery & Preservation

- Paradigm Shift
  - New technologies moving away from cumbersome capture/identification processes
  - Growing role in civil litigation, sometimes extending to criminal/tribunal cases
- Challenges
  - High cost of forensic experts + legal analyst teams
  - Keyword searches → slow, overwhelming, limited by human review capacity
  - Large datasets hinder meticulous evidence examination
- Processes
  - No forensic imaging required in e-discovery workflows
- Automation vs Manual Review
  - Automated tools save costs but face skepticism vs “gold standard” manual review
  - Debate continues on accuracy and completeness of automated methods
- Indexing
  - Software scans text, builds searchable databases. Optimizes retrieval for repositories, archives, and business records

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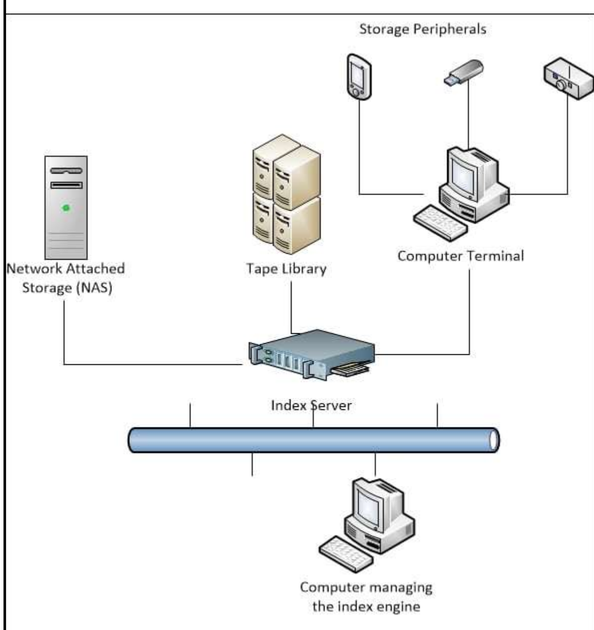

Typically, in index-based and non-index-based e-discovery data collection, the software is installed on the client server or network, which permits the inspection and indexing of the selected corpus, which is held as a searchable on an agent or index server for later retrieval and processing

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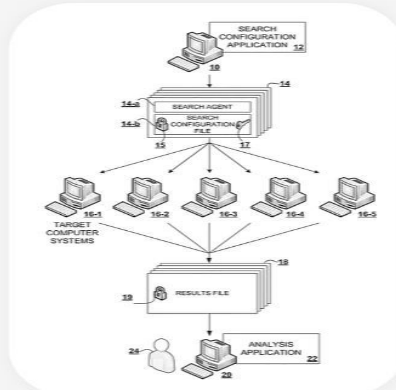
XTREMEFORENSICS

ISEEK
THE INTELLIGEN
EDISCOVERY - DIGITAL FORENSICS - MAL

ISEEK KEY FEATURES

- Runs without the need for indexing
- Searches across all drives and network shares
- No installation required
- No dongles involved
- Defensible and verifiable data collection
- Fast search engine featuring parallel processing
- Searches live/locked documents including email, compressed files and all document types
- Produces 256 AES encrypted, password-protected output stores
- Export responsive content to review tools
- Automatic file inclusions and exclusions
- Captures metadata on all files collected
- The requirement for De-Nisting is removed
- Covert operation



THE ISEEK PATENTED PROCESS

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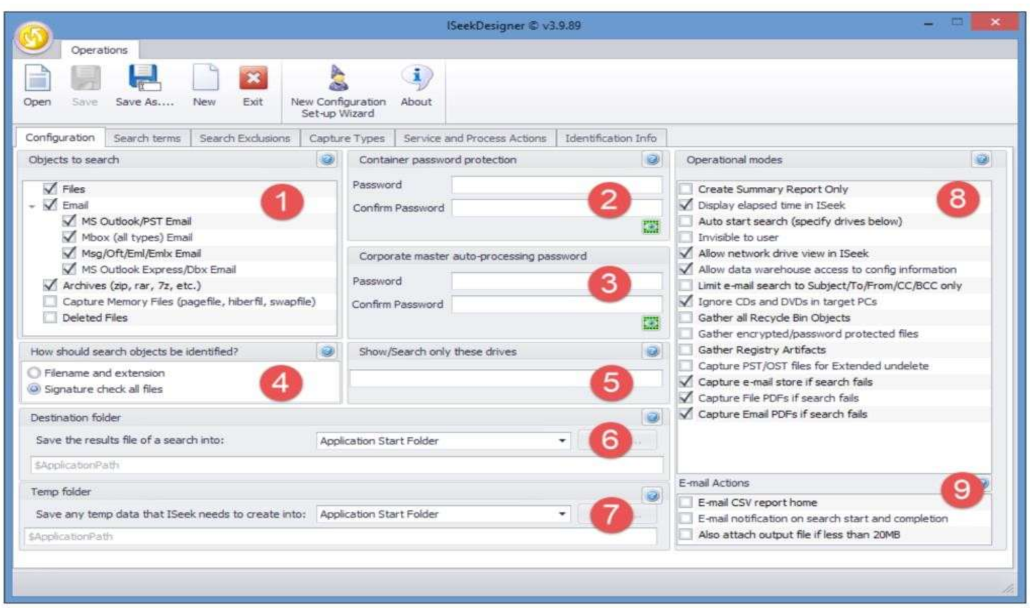

The opening pane of the ISeekDesigner help wizard (Configuration)

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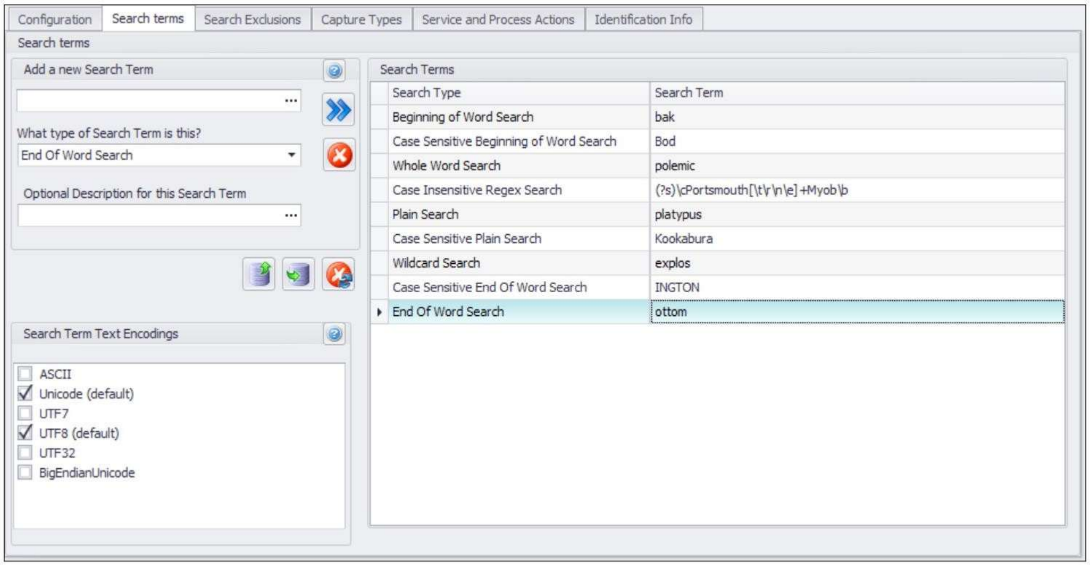

The opening pane of the ISeekDesigner help wizard (Search terms)

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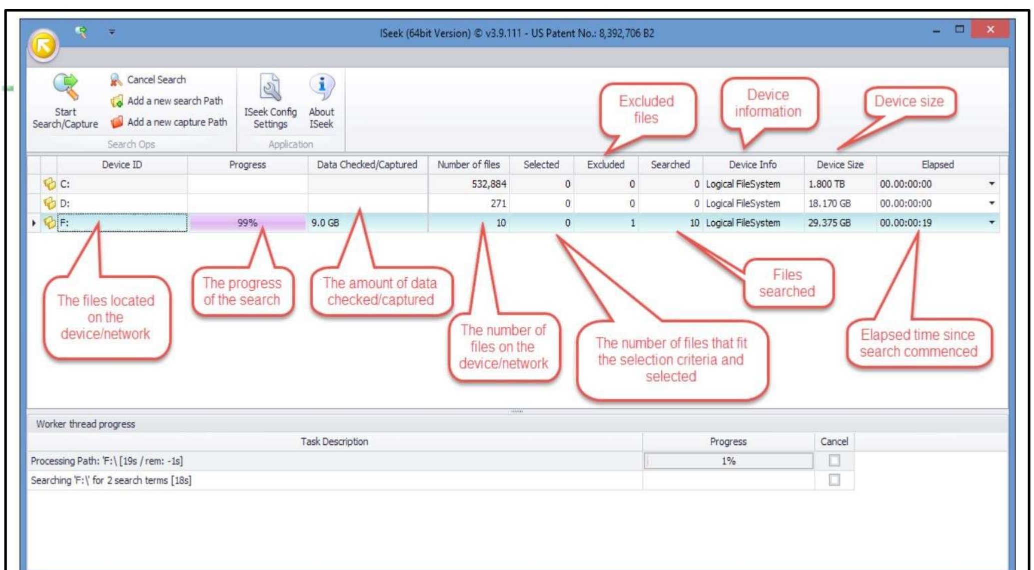

Viewing the ISeekDiscovery automaton searching a selected computer drive

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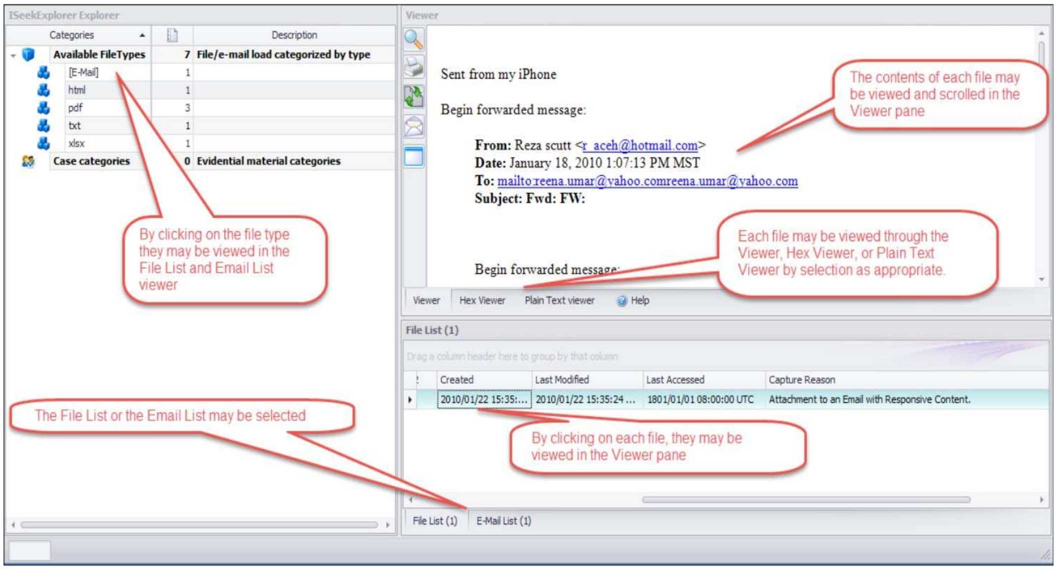

The ISeekExplorer view of the files captured and file content and metadata

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Software Auditing as E-Discovery

- Many commercial software distributors have methods to perform software auditing/forensics on live workstations/PCs to identify software licensing fraud.

Adobe License Audit(1072)

Domain	AssetName	Type	Asset state	Lastseen	OS	SoftwareName	SoftwarePublisher	SoftwareVersion
Lansweeper	XLAN-431	Desktop	Active	04/05/2012 16:36:29	Microsoft Windows XP Professional	Adobe Photoshop CS	Adobe Systems, Inc.	CS
Lansweeper	XLAN-359	Desktop	Active	18/06/2015 08:09:00	Microsoft Windows 7 Enterprise	Adobe AcrobatPro 9.5.4 Gen P0	Adobe Systems	9.5.4
Lansweeper	XLAN-8	Desktop	Active	21/02/2013 12:25:42		Adobe Acrobat 9 Pro	Adobe Systems	9.5.3
Lansweeper	XLAN-210	Laptop	Active	07/09/2012 07:11:26	Microsoft Windows 7 Professional	Adobe Reader 9.5.2	Adobe Systems Incorporated	9.5.2
Lansweeper	XLAN-424	Server	Active	04/09/2012 09:29:53	Microsoft(R) Windows(R) Server 2003, Enterprise Edition	Adobe Reader 9.5.2	Adobe Systems Incorporated	9.5.2
Lansweeper	XLAN-135	Laptop	Active	07/09/2012 08:31:11	Microsoft Windows XP Professional	Adobe Reader 9.5.1	Adobe Systems Incorporated	9.5.1
Lansweeper	XLAN-138	Laptop	Active	07/09/2012 10:18:58	Microsoft Windows XP Professional	Adobe Reader 9.5.1	Adobe Systems Incorporated	9.5.1
Lansweeper	XLAN-327	Desktop	Active	24/06/2013 11:49:50	Microsoft Windows XP Professional	Adobe Reader 9.5.0	Adobe Systems Incorporated	9.5.0
Lansweeper	XLAN-357	Desktop	Active	12/01/2012 13:26:21	Microsoft® Windows Vista™ Enterprise	Adobe Reader 9.4.0 Generic R0	Adobe Systems Incorporated	9.4.7
Lansweeper	XLAN-361	Laptop	Active	12/01/2012 12:00:57	Microsoft® Windows Vista™ Enterprise	Adobe Reader 9.4.0 Generic R0	Adobe Systems Incorporated	9.4.7
Lansweeper	XLAN-418	Desktop	Active	04/09/2012 09:08:07	Microsoft Windows 7 Ultimate	Adobe Reader 9.4.7	Adobe Systems Incorporated	9.4.7
Lansweeper	XLAN-145	Laptop	Active	07/09/2012 09:16:17	Microsoft Windows XP Professional	Adobe Reader 9.4.6	Adobe Systems Incorporated	9.4.6
Lansweeper	XLAN-198	Laptop	Active	07/09/2012 08:01:45	Microsoft Windows XP Professional	Adobe Reader 9.4.6	Adobe Systems Incorporated	9.4.6
Lansweeper	XLAN-313	Desktop	Active	23/05/2012 15:00:10	Microsoft Windows 7 Enterprise	Adobe Reader 9.4.6 MUI	Adobe Systems Incorporated	9.4.6
Lansweeper	XLAN-352	Desktop	Active	06/11/2012 15:08:32	Microsoft Windows XP Professional	Adobe Reader 9.4.5	Adobe Systems Incorporated	9.4.5
Lansweeper	XLAN-355	Desktop	Active	07/11/2012 13:35:00	Microsoft Windows XP Professional	Adobe Reader 9.4.5	Adobe Systems Incorporated	9.4.5
Lansweeper	XLAN-315	Desktop	Active	11/03/2016 16:32:59	Microsoft Windows 7 Professional	Adobe Acrobat 9 Pro - Italiano, Español, Nederlands, Português	Adobe Systems	9.4.4

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An Example Relevant To YOU!

Vault

Data retention and eDiscovery for Google Workspace.

What is eDiscovery?

Electronic discovery, or eDiscovery, is the process of seeking and finding information in electronic format. It is typically used in response to legal matters and investigations.

What happens to data in deleted accounts?

If you delete a user, all data associated with the user's account will be removed from Google. As best practice, Google recommends suspending user accounts instead of deleting them.

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Enhanced Digital Evidence Recovery & Preservation

Some benefits include

- Indexing without contamination to produce solid evidence identification
- A significant reduction in time required to complete data capture via indexing versus searching → also in the amount of data requiring capture
- Avoidance of site visits and associated travel
- A "safe harbor" for captured data that can be transported speedily and at lower cost
- No contamination of the evidence collated
- Simplicity of access to the target datasets without an Internet connection or software installation
- Simple executables with a minimum of technical expertise required
- Customizable search options compatible with analyst and legal team objectives

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E-Discovery vs Digital Forensics

Aspect	E-Discovery	Digital Forensics
Purpose	Collect and produce electronic data for legal cases.	Investigate and analyze digital devices for evidence.
Main Focus	Finding relevant ESI (emails, docs).	Recovering, preserving, and examining digital evidence.
Process Type	High-volume data search and review.	Technical, detailed forensic examination.
Data Sources	Emails, servers, cloud data.	Hard drives, phones, USBs, logs.
Tools	Relativity, Nuix.	EnCase, FTK, Autopsy, Cellebrite.
Integrity Level	Metadata may change.	Strict preservation, hashing, chain of custody.
Used In	Civil litigation.	Criminal investigations, cyber incidents.

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Empowering First Responders at Crime Scenes

- As noted in earlier lectures, a first responder (e.g. network admin, IT manager, first on scene police officer, etc) may have good intentions when trying to determine whether a transgression has occurred and are attempting to preserve evidence → can amount to unintentional evidence tampering if they do not any forensic experience and/or the right tools
- It is long overdue that some form of basic training and automated tools be able to assist these stakeholders in managing the identification and collection of potential evidence without contamination → to have forensic practitioners focus solely on evidence recovery and discovery.
- Law enforcement field agents can be tasked to be evidence collectors
- Whether or how they do it is a judgment call for the officer at the scene
- The big issue here is the variety of devices available out there and not every agent may know how to handle them properly

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Challenges in Crime Scences

- **Detecting Incidents:** Difficulty knowing when a breach or misconduct has occurred (e.g., hacker attacks, insider threats, staff misbehavior).
- **Identifying Suspicious Activity:** Hidden intentions, missing personnel, confidentiality breaches, audit anomalies, or fraud indicators.
- **Role of IT Custodians:** They understand system data, access patterns, and applications often becoming the first responders.
- **Evidence Requirements:** Investigations may need incriminating evidence, but deeper forensic tools/expertise may be lacking.
- **Complex Data Environments:** Evidence stored on network servers complicates collection without disrupting operations.
- **Data Volume Explosion:** Growing storage sizes make full imaging impractical necessitating live forensics and indexing tools.

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Summary

What was covered today:

- We looked at evidence recovery and preservation in a general sense and then focused on how these two interlinked requirements relate to digital evidence
- Dead and live evidence recovery processes have been described, although these well-established processes are now facing a paradigm shift heralded by enhanced technologies

Reference Source: Practical Digital Forensics (Packt Publishing)

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THANKS – Q&A

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